PIGMENTED LESIONS OF ORAL MUCOSA

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LECTURE LEARNING OUTCOMES

At the end of the lecture, students should be able to:

Classify the pigmented lesions of oral cavity

Summaries the terminology related to oral pigmented lesions

Discuss melanin and metal associated oral pigmentation

Describe Oral nevus and Malignant Melanoma

Essential reading:

Regezi & Sciubba: Oral Pathology - Clinicopathologic Correlation; r (6th edn), Chapter 5, pg: 134-148

Shafers textbook oral pathology((7thedn)

INTRODUCTION

- Human oral mucosa is not uniformly colored
- Physiologic and pathologic conditions

➢ **Degree of keratinization**

➢ **Numbers and melanogenic activity of melanocytes**

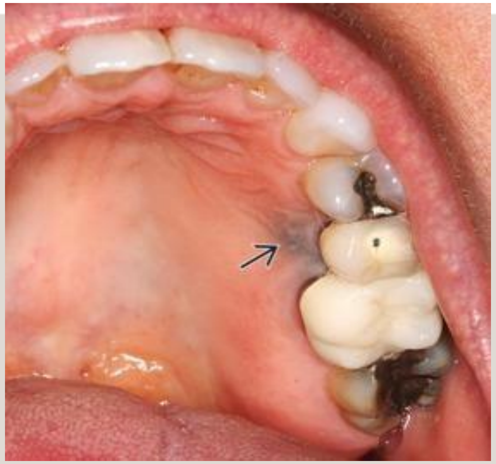
➢ **Vascularization**

Physiological and Pathological Pigmentation

PIGMENT

Color or Coloring Agent

- “Pigmentation is defined as the process of discoloration of the oral mucosa or gingiva due to accumulation of pigments”.



- **Endogenous:** melanin and blood-related pigments (Hemoglobin/Bilirubin/Hemolysis).
- **Exogenous**: metals and drug-related pigments.

CLASSIFICATION OF PIGMENTED LESIONS

Pigmented lesions associated with melanin (BROWN)

Smoker's melanosis Oral

Melanotic macule

Café au lait pigmentation

Oral melanoacantthoma

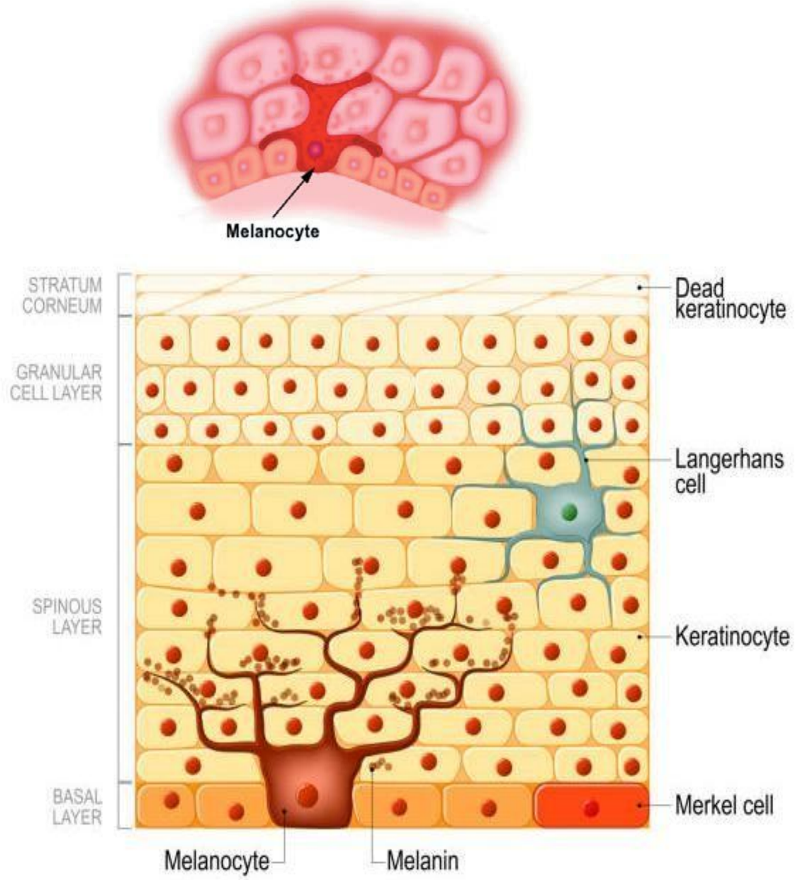
Melanocytic nevus

Malignant melanoma

Drug induced melanosis

Inflammation associated hyperpigmentation

Racial pigmentation



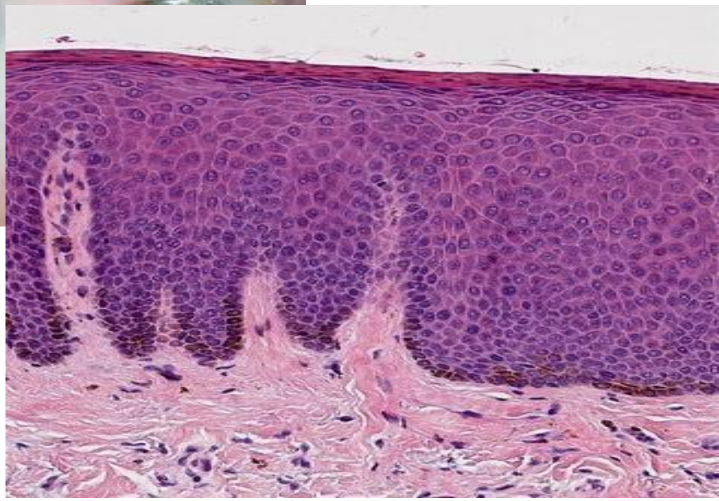
Blue/purple vascular lesions

1. Hemangioma
2. Varix
3. Angiosarcoma
4. Kaposi's sarcoma

Physiologic pigmentation: (Racial)

- Black people and Asians show diffuse melanosis of facial gingiva
- Lingual gingiva and tongue may exhibit multiple, diffuse and reticulated brown macules
- Treatment: gingivectomy and laser therapy

TOBACCO ASSOCIATED MELANOSIS (SMOKERS MELANOSIS)



TOBACCO ASSOCIATED MELANOSIS (SMOKERS MELANOSIS)

- Abnormal melanin pigmentation of oral mucosa has been linked to cigarette smoking.

- Mechanism :

1. A component in tobacco smoke that stimulates melanocytes
2. Hormones also play a role in SAM.
3. Heat also related to pigmentation.

CAFÉ-AU-LAIT MACULE

- Cafe-au-lait (coffee with milk.) macules are discrete melanin-pigmented patches of skin that have irregular margins and a brown coloration.
- They are noted at birth or soon thereafter and may also be seen in normal children.
- No treatment is required, but they may be indicative of a Syndrome.
- **McCune –Albright syndrome:** **polyostotic Fibrous dysplasia,** various endocrinopathies, soft tissue myxomas
- Individuals with **six or more café-au-lait macules larger than 1.5 cm in diameter** (in post pubertal individuals) — or **larger than 0.5 cm in prepubertal individuals** — should be suspected of having **Neurofibromatosis Type 1 (NF1).**

ORAL MELANOTIC MACULE

- **Most common oral mucosal lesions** of melanocytic origin.
- **Lower lip** and **gingival (ant.Maxilla)**
  Homogeneously colored and less than 1 cm in diameter.
- Caused by an **increased production and deposition of melanin within the basal cell layer**
- Syndrome associated: **Peutz Jeghers syndrome.**

- Intra oral freckle

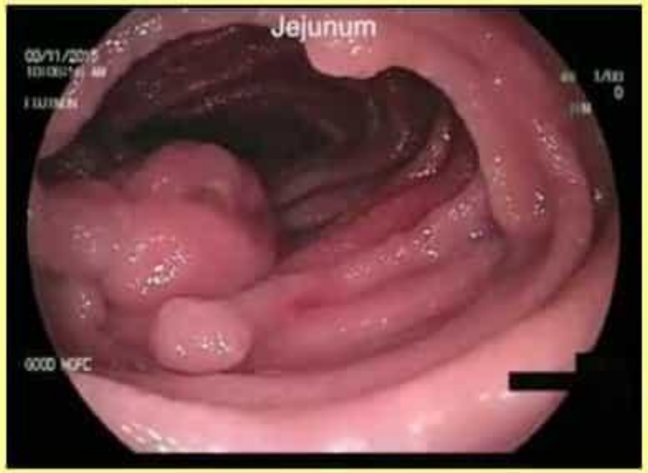
Peutz-Jeghers syndrome

- Is an **autosomal dominant** disease
- **Intestinal polyposis**, cancer susceptibility and multiple, small, **pigmented macules of lips**, **perioral skin, hands and feet**
- Macules, less than 0.5 cm in diameter
- These may appear on anterior tongue and buccal and labial mucosa
- **Lip and perioral region involvement is pathognomonic**

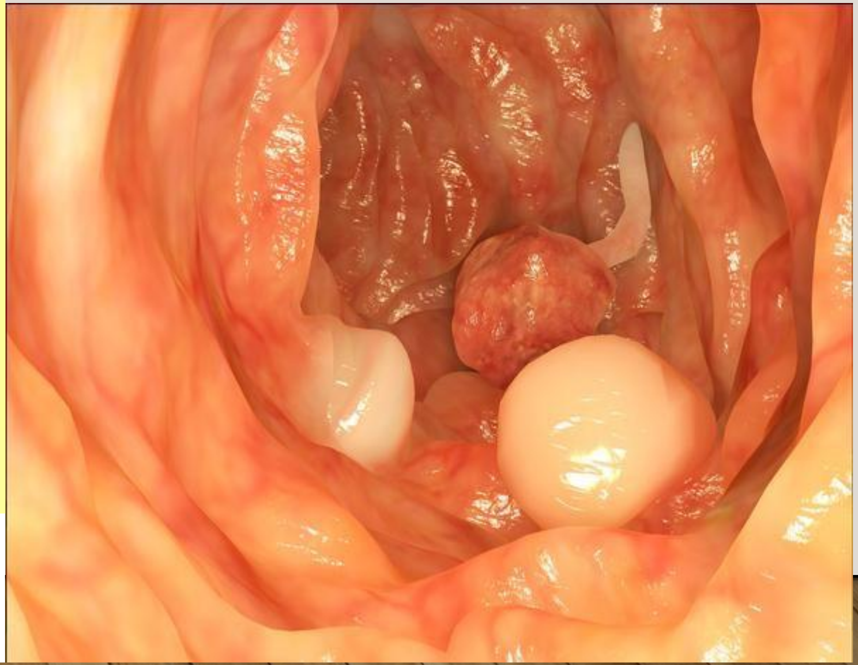
Melanosis on lips

Melanosis in oral mucosa

Melanosis on finger



Hamartomatous polyps of small intestine



Peutz Jeghers syndrome

AMALGAM TATTOO

- Most common pigmentation in oral cavity
- Iatrogenic lesion – Direct or friction with restoration
- Soluable silver compounds responsible for the color
- Macular lesions and grey in color
- Gingiva, buccal mucosa, palate and tongue are common sites
- Occasionally detected in radiographs if size of Amalgum is large

Drug induced melanosis

- A variety of drugs can induce oral pigmentation
- Lesions are flat and no evidence of nodularity or swelling
- **Quinolone and hydroxyquinolone antimalarials, cyclophosphamide**

- **Minocycline** used for treatment of acne.
- Discoloration tends to fade within a few months after the drug is discontinued

Post inflammatory hyper pigmentation

- Present as **focal or diffuse pigmentation of area subject to previous injury or inflammation**
- Acne prone face is relatively common site
- This type of pigmentation is seen in patient with Lichen planus

MELANOCYTIC NEVI

- Congenital papular/nodular lesion madeup of melanocytic cells.
- Nevocullar nevus or pigmented nevus.
- Nevus cells are seen in epithelium or connective tissues.
- Origin from neural creast or modifications of melanocytes.

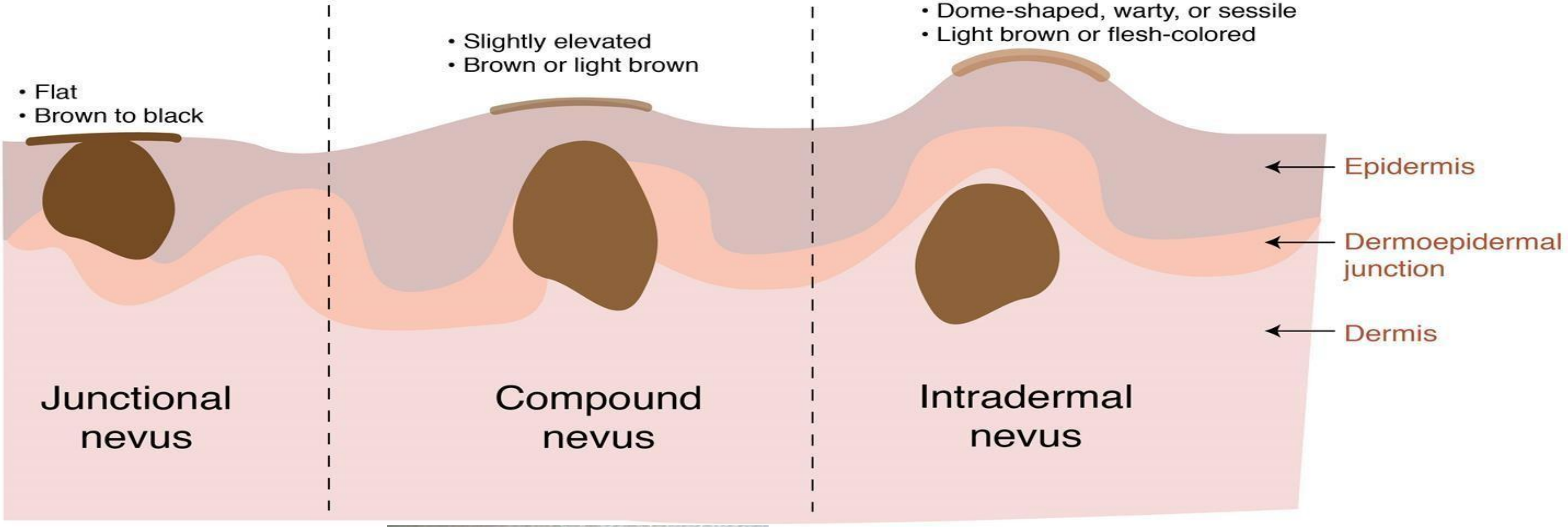
ORAL NEVI

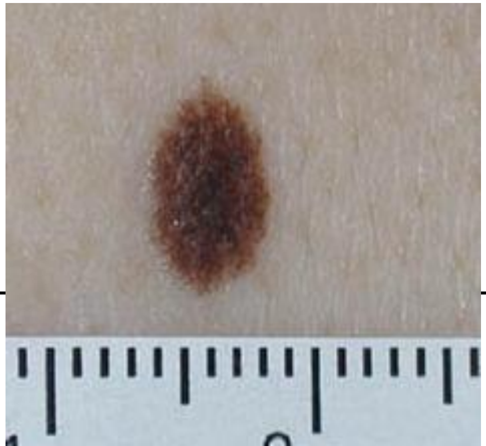
- Intraoral MN are uncommon
- Younger than 40 years
- Common site - HARD PALATE
- Most of them **asymptomatic**

COMMON IN OCCURRENCE

1. Intramucosal
2. Blue
3. Compound
4. Junctional

Melanocytic Nevi





HISTOPATHOLOGY

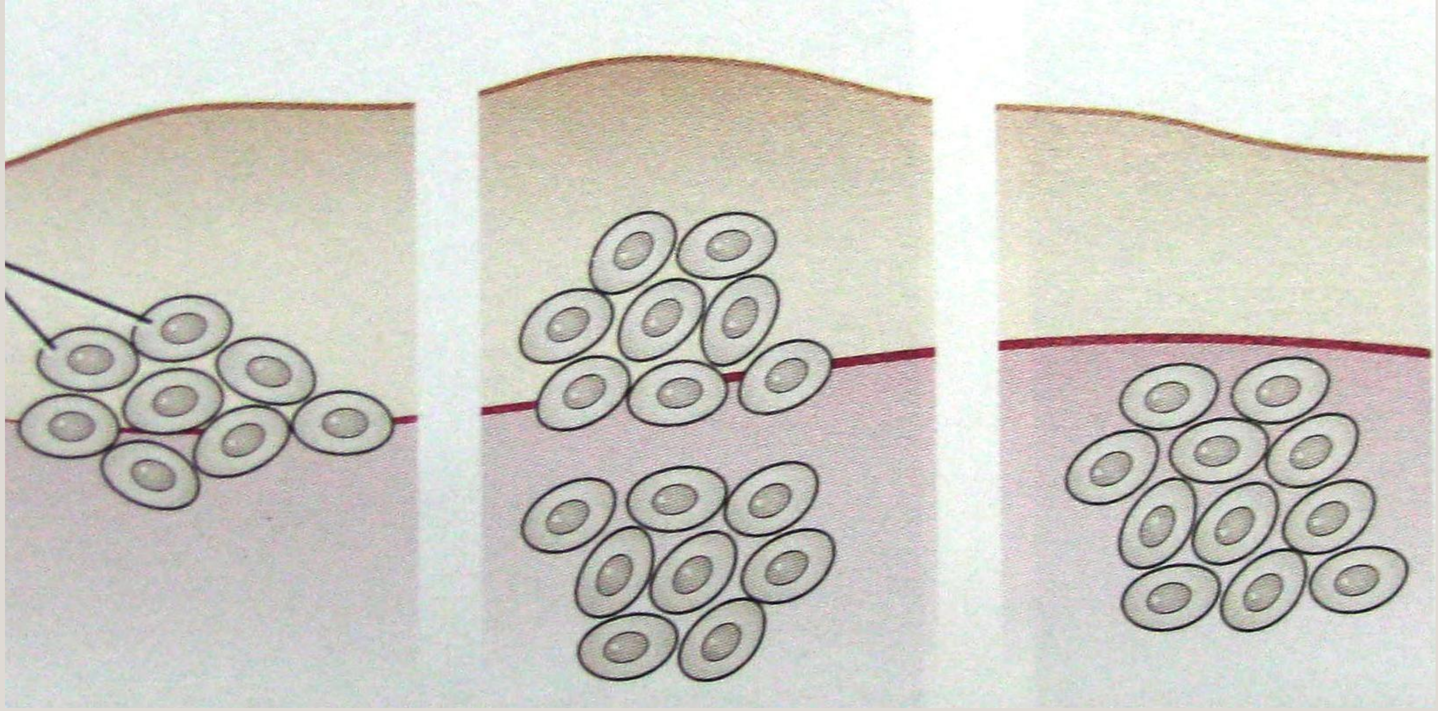
Subtypes are depending up on the location of nevus cells.

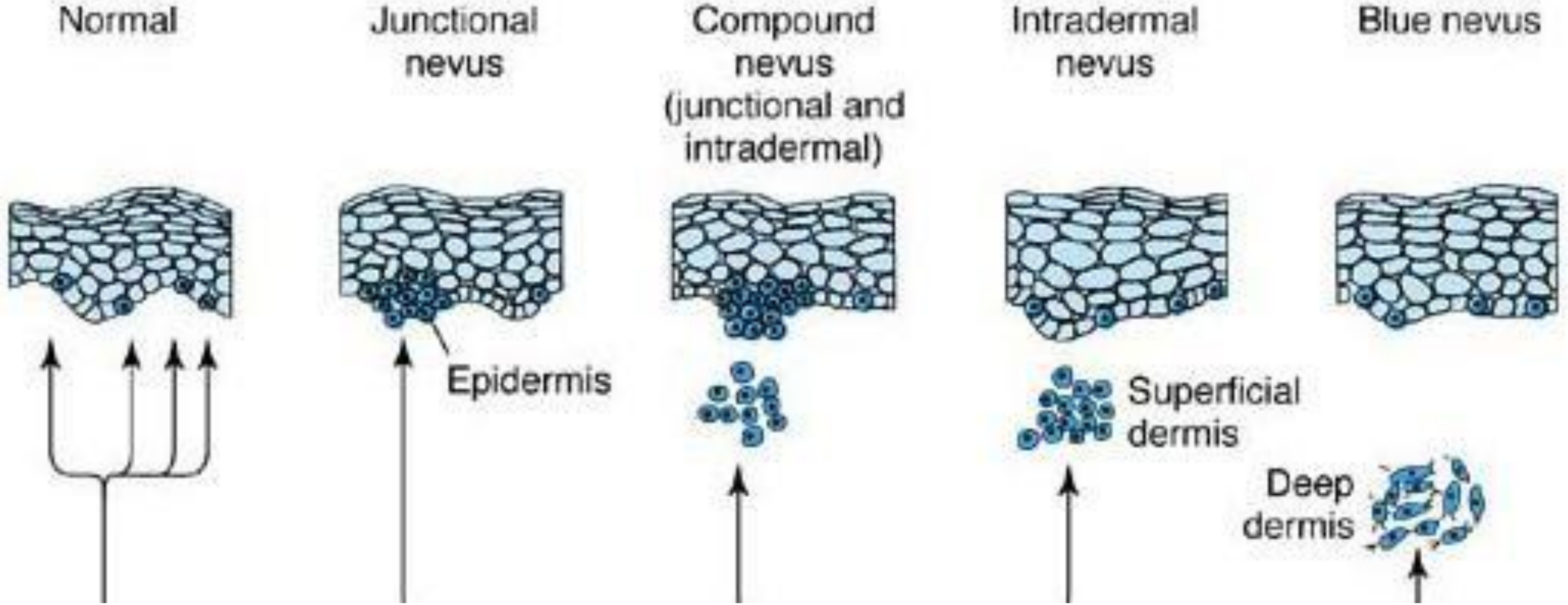
1. **Intradermal nevus or intramucosal nevus**
2. **Junctional nevus**
3. **Compound nevus**

A fourth type of nevus - spindle shaped cells - deep in the CT

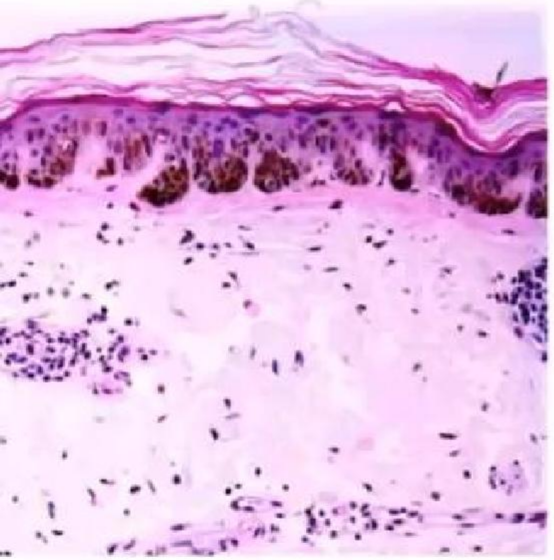
BLUE NEVUS

- Melanocytic nevi are collections of nevus cells that are round or polygonal and are typically seen in a nested pattern.
- They may be found in epithelium or supporting connective tissue, or both.

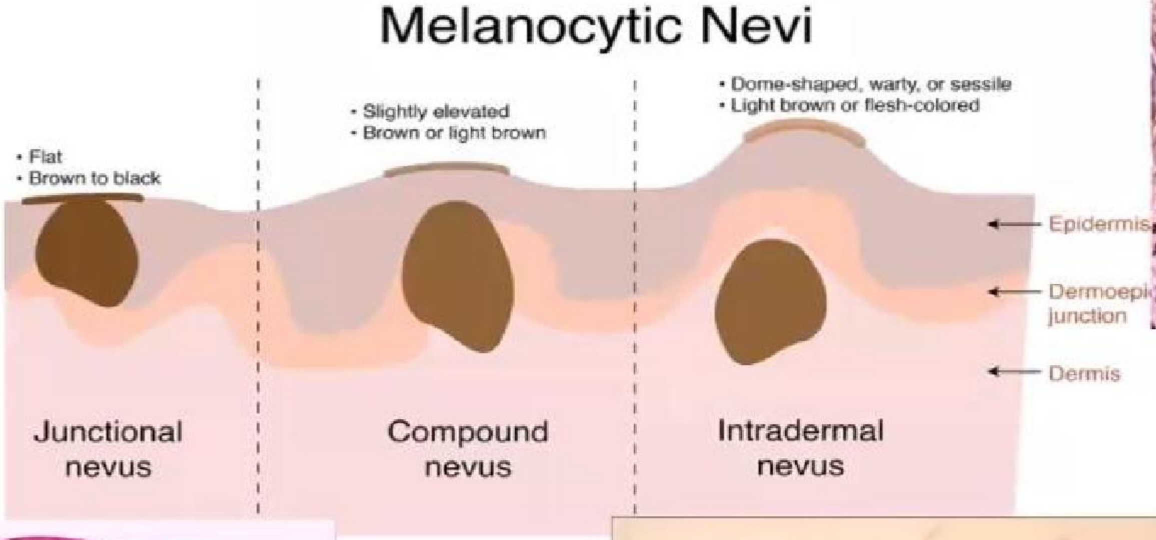


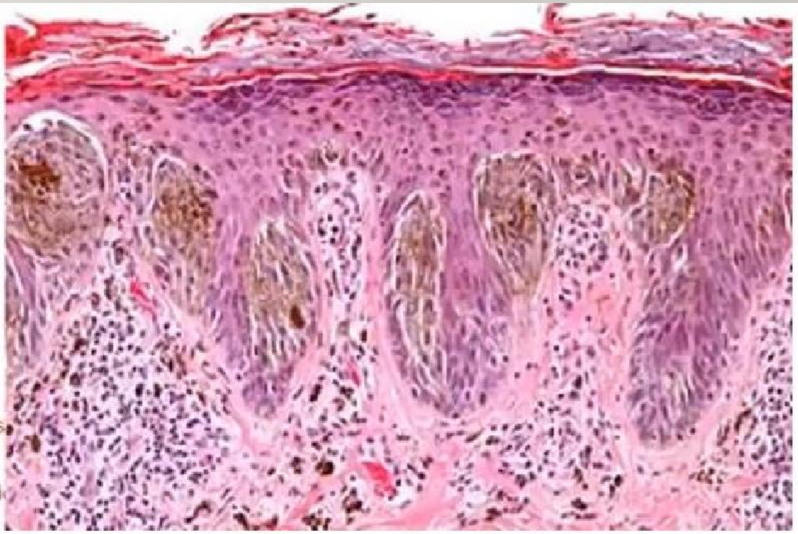

Normal
Junctional nevus
Compound nevus (junctional and intradermal)
Intradermal nevus
Blue nevus
Epidermis
Superficial dermis
Deep dermis

Histopathology

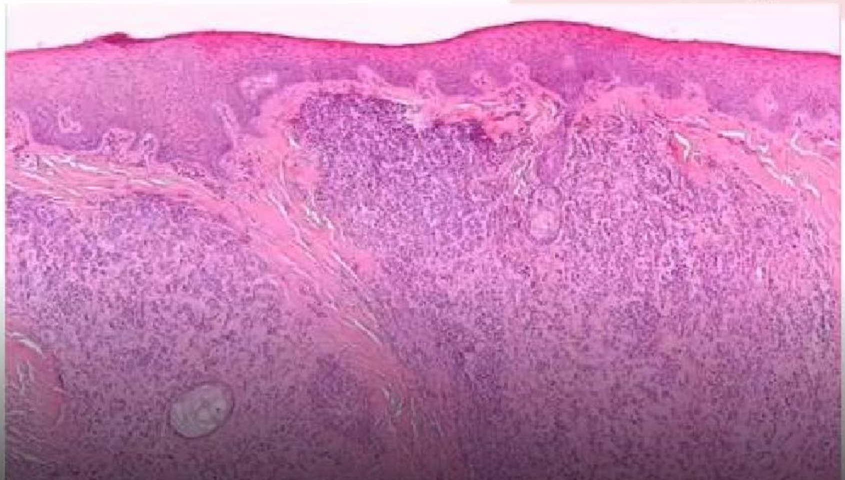


Junctional

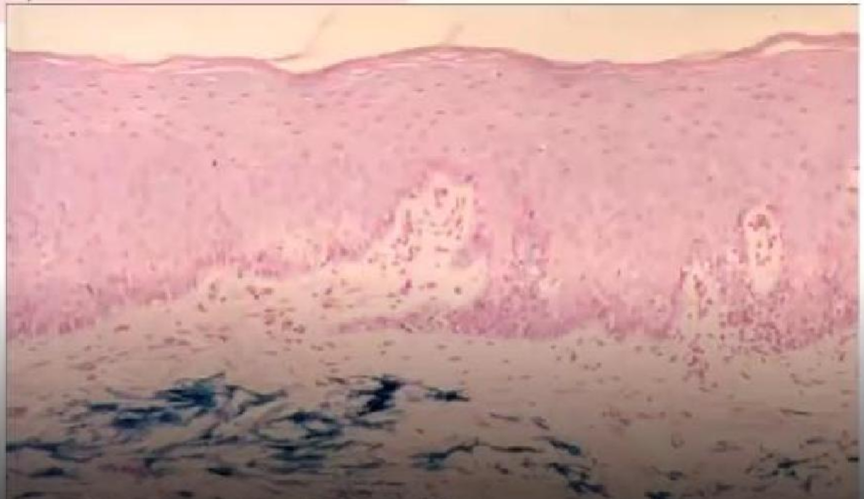




Compound



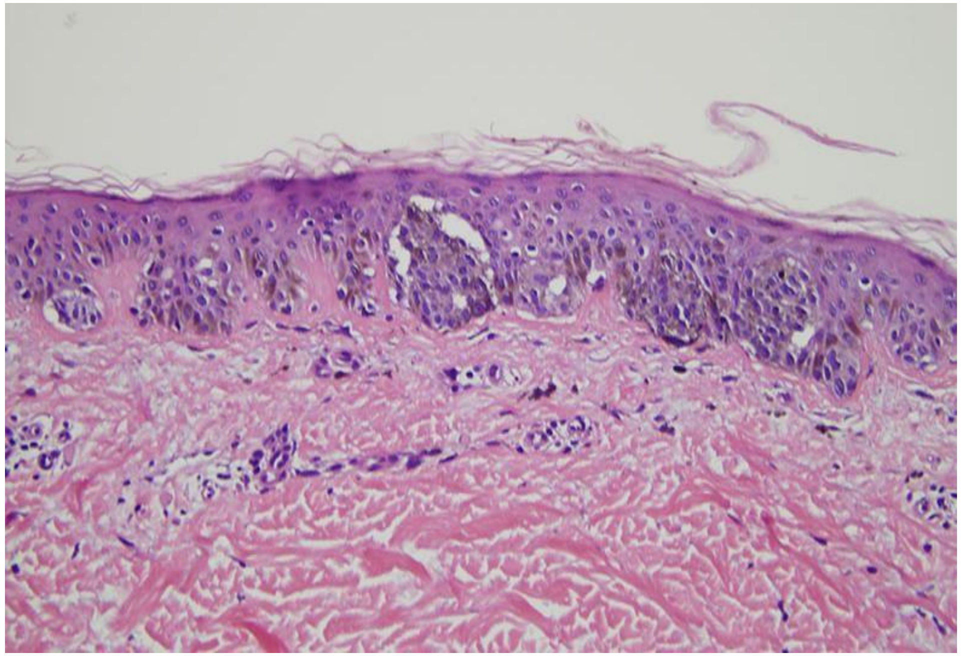
Intradermal



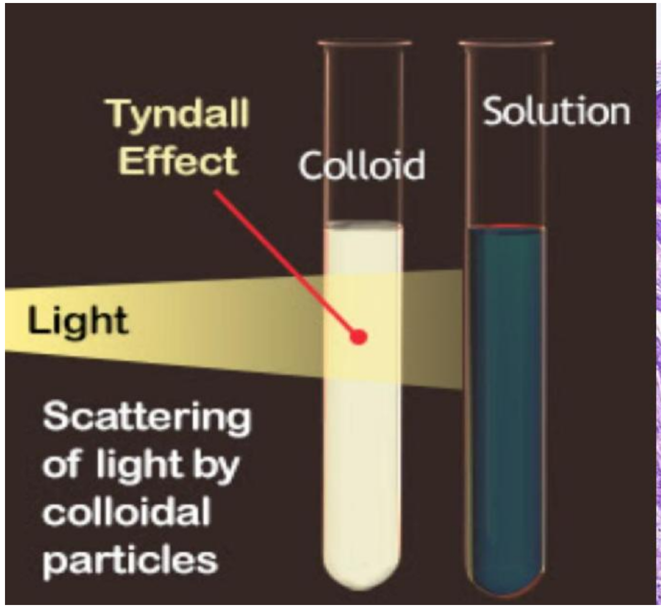
Blue Nevus

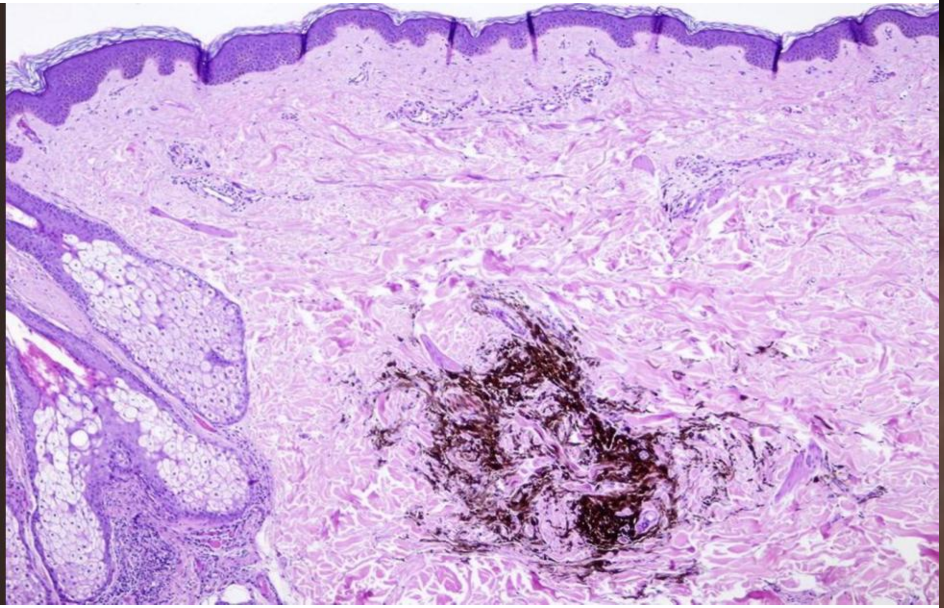
Junctional Nevus

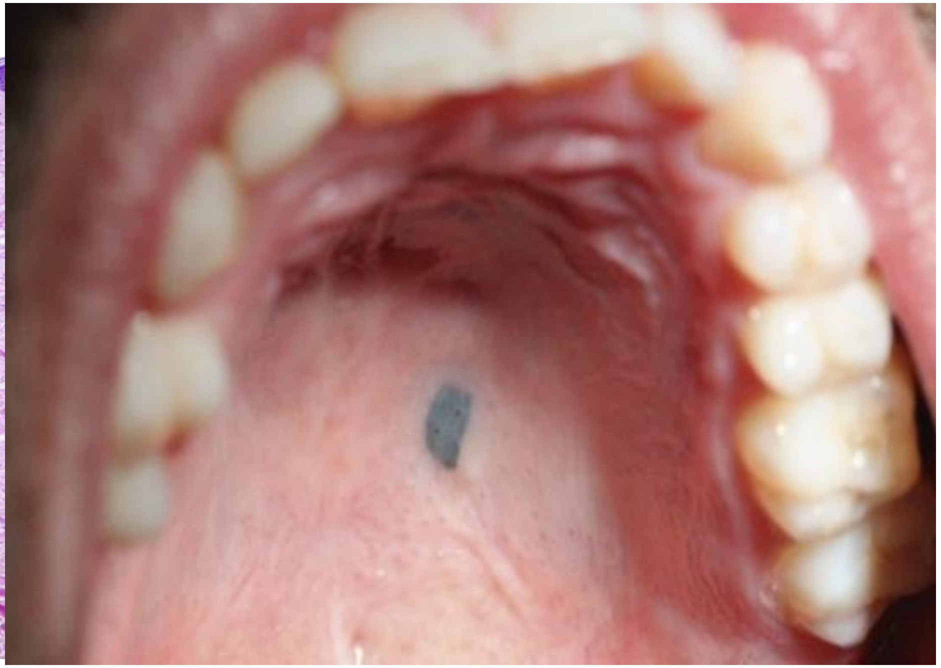
- Nevus cells contact and blend into epithelium
- Overlying epithelium thin and irregular
- Shows cells are crossing junction and growing down into CT.**(Dropping off effect)**
- This junctional activity has serious implications: **high rate of malignant transformation**



BLUE NEVUS


Tyndall Effect
Colloid
Solution
Light
Scattering of light by colloidal particles

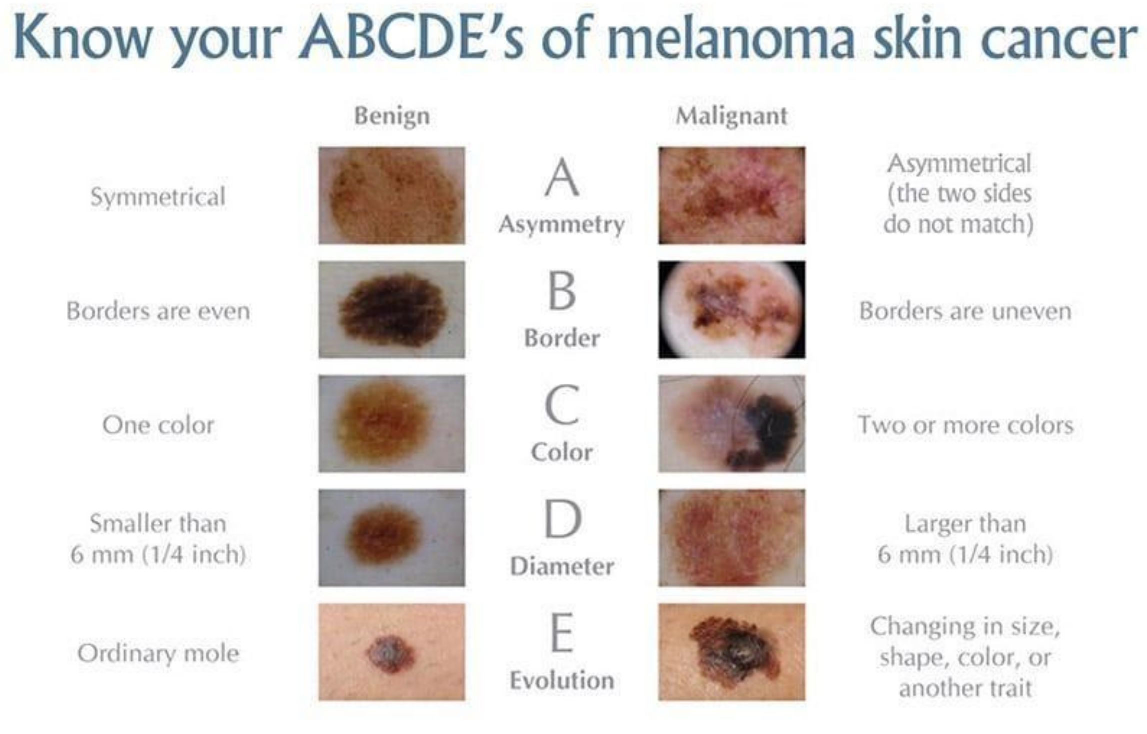




TREATMENT

A B C D E RULE

- No treatment is required, unless it is clinically indicated,
- If clinically mimic oral melanoma than biopsy is advisable



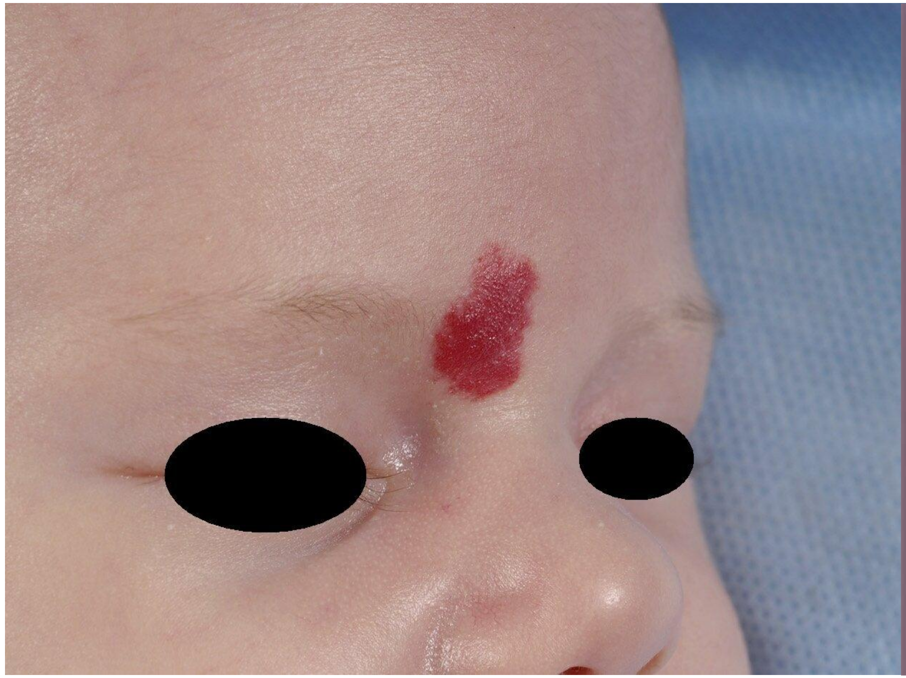
Blue/purple vascular lesions

- Hemangioma
- Varix
- Angiosarcoma
- Kaposi’s sarcoma

Hemangioma

- Hemangioma is a **benign vascular tumour characterised by proliferation of endothelial cells forming a mass of blood vessels.**
- In the oral and maxillofacial region, it most commonly appears on lips, tongue, buccal mucosa
- Typical clinical features: a **soft, compressible, bluish-reddish swelling** which may b**lanch on pressure (diascopy)** and can enlarge with trauma.
- **Clinical Significance** : Lesions may pose a risk of significant bleeding during dental extractions or surgical procedures due to their vascular nature

- Many superficial lesions may involute spontaneously, especially in infancy.
- In the oral region, treatment includes sclerotherapy, laser therapy, surgical excision




a
b
c
d

Varix

- A **focal dilatation of a vein** due to loss of vascular wall elasticity and weakened venous support tissues.
- Considered an **acquired vascular lesion**, not a true tumor (unlike hemangioma).
- Occurs due to age-related degeneration of elastic fibers in the vein wall.
- May result from chronic venous stasis, increased venous pressure, or trauma.
- Ventral surface of tongue (most frequent – “lingual varices”).Lower lip, buccal mucosa, and floor of mouth.

Clinical features:

- Bluish-purple, compressible, soft papule or nodule.
- Size: few millimeters to 1 cm.
- Blanches on pressure (positive diascopy).
- Usually asymptomatic

Diagnosis : Clinical examination + diascopy (blanching test).

Treatment: Usually, no treatment required unless for esthetic or functional reasons.

If needed: sclerotherapy, laser ablation, or surgical excision.


Thank you